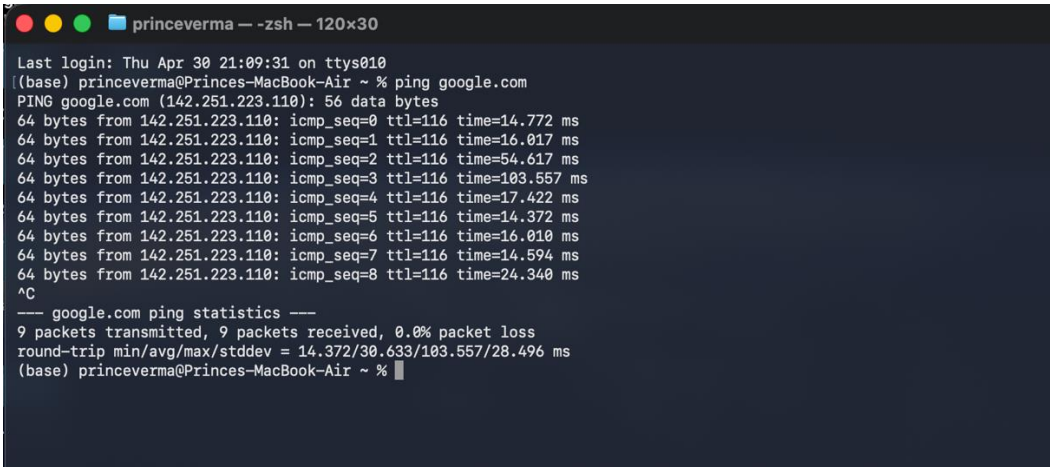
CN ASSIGNMENT -2

1. To capture and analyze ICMP packets (Ping request and reply)



```
princeverma — -zsh — 120×30
Last login: Thu Apr 30 21:09:31 on ttys010
(base) princeverma@Princes-MacBook-Air ~ % ping google.com
PING google.com (142.251.223.110): 56 data bytes
64 bytes from 142.251.223.110: icmp_seq=0 ttl=116 time=14.772 ms
64 bytes from 142.251.223.110: icmp_seq=1 ttl=116 time=16.017 ms
64 bytes from 142.251.223.110: icmp_seq=2 ttl=116 time=54.617 ms
64 bytes from 142.251.223.110: icmp_seq=3 ttl=116 time=103.557 ms
64 bytes from 142.251.223.110: icmp_seq=4 ttl=116 time=17.422 ms
64 bytes from 142.251.223.110: icmp_seq=5 ttl=116 time=14.372 ms
64 bytes from 142.251.223.110: icmp_seq=6 ttl=116 time=16.010 ms
64 bytes from 142.251.223.110: icmp_seq=7 ttl=116 time=14.594 ms
64 bytes from 142.251.223.110: icmp_seq=8 ttl=116 time=24.340 ms
^C
--- google.com ping statistics ---
9 packets transmitted, 9 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 14.372/30.633/103.557/28.496 ms
(base) princeverma@Princes-MacBook-Air ~ %
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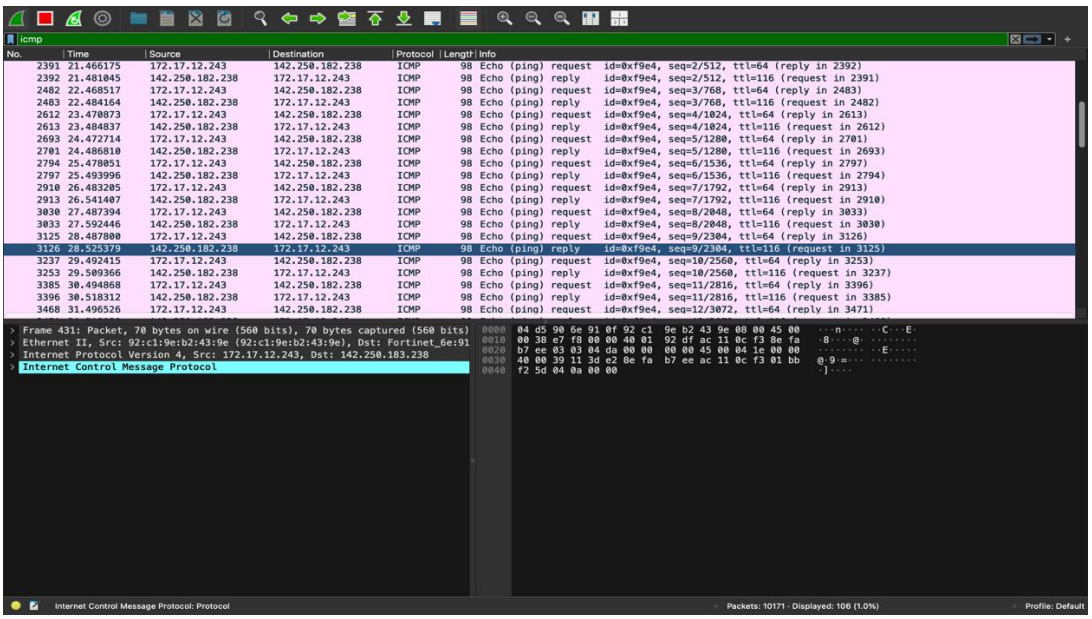

icmp

No.	Time	Source	Destination	Protocol	Length	Info
2391	21.466175	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=2/512, ttl=64 (reply in 2392)
2392	21.481045	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=2/512, ttl=116 (request in 2391)
2482	22.468517	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=3/768, ttl=64 (reply in 2483)
2483	22.484164	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=3/768, ttl=116 (request in 2482)
2612	23.470873	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=4/1024, ttl=64 (reply in 2613)
2613	23.484837	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=4/1024, ttl=116 (request in 2612)
2693	24.472714	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=5/1280, ttl=64 (reply in 2701)
2701	24.486810	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=5/1280, ttl=116 (request in 2693)
2794	25.478051	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=6/1536, ttl=64 (reply in 2797)
2797	25.493996	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=6/1536, ttl=116 (request in 2794)
2910	26.483205	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=7/1792, ttl=64 (reply in 2913)
2913	26.541407	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=7/1792, ttl=116 (request in 2910)
3030	27.487394	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=8/2048, ttl=64 (reply in 3033)
3033	27.592446	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=8/2048, ttl=116 (request in 3030)
3125	28.487800	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=9/2304, ttl=64 (reply in 3126)
3126	28.535370	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=9/2304, ttl=116 (request in 3125)
3237	29.492415	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=10/2560, ttl=64 (reply in 3253)
3253	29.509366	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=10/2560, ttl=116 (request in 3237)
3385	30.494868	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=11/2816, ttl=64 (reply in 3396)
3396	30.518312	142.250.182.238	172.17.12.243	ICMP	98	Echo (ping) reply id=0xf9e4, seq=11/2816, ttl=116 (request in 3385)
3468	31.496526	172.17.12.243	142.250.182.238	ICMP	98	Echo (ping) request id=0xf9e4, seq=12/3072, ttl=64 (reply in 3471)

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> Frame 431: Packet, 70 bytes on wire (560 bits), 70 bytes captured (560 bits)
> Ethernet II, Src: 92:c1:9e:b2:43:9e (92:c1:9e:b2:43:9e), Dst: Fortinet_6e:91
> Internet Protocol Version 4, Src: 172.17.12.243, Dst: 142.250.183.238
> Internet Control Message Protocol

0000  04 d5 90 6e 91 0f 92 c1  9e b2 43 9e 08 00 45 00   ···n···· ··C···E·
0010  00 38 e7 f8 00 00 40 01  92 df ac 11 0c f3 8e fa   ·8····@· ····· ····
0020  b7 ee 03 03 04 da 00 00  00 00 45 00 04 1e 00 00   ····· ··E·····
0030  40 00 39 11 3d e2 8e fa  b7 ee ac 11 0c f3 01 bb   @·9·=··· ····· ····
0040  f2 5d 04 0a 00 00                                  ·]····
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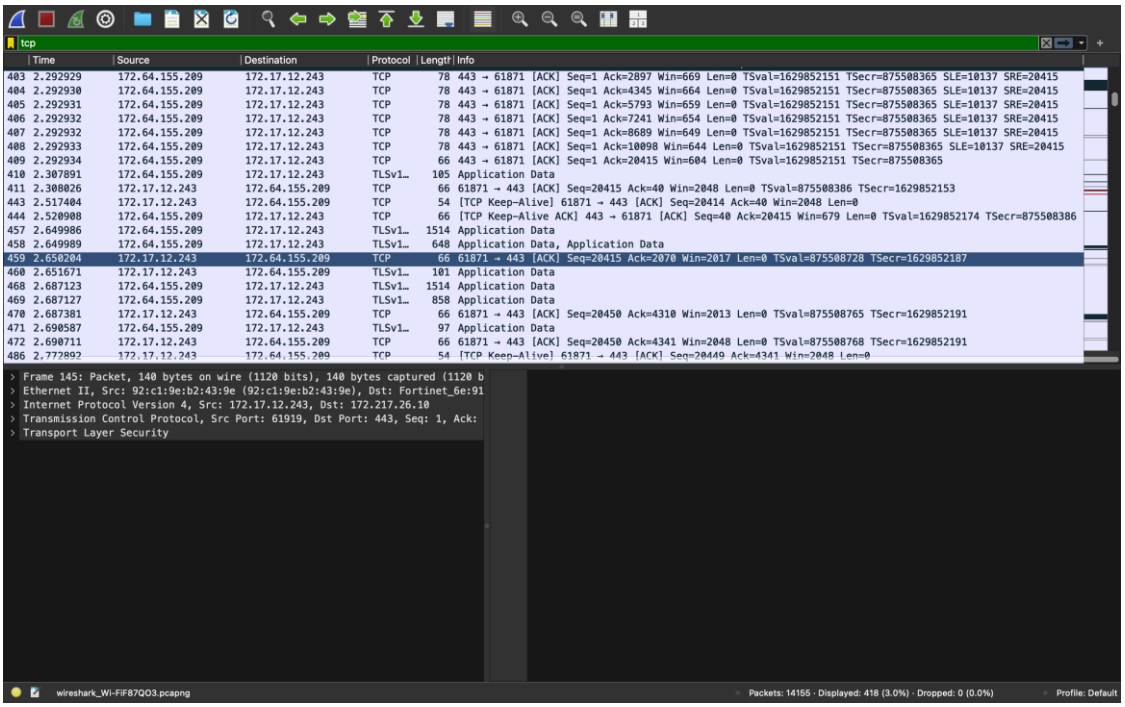
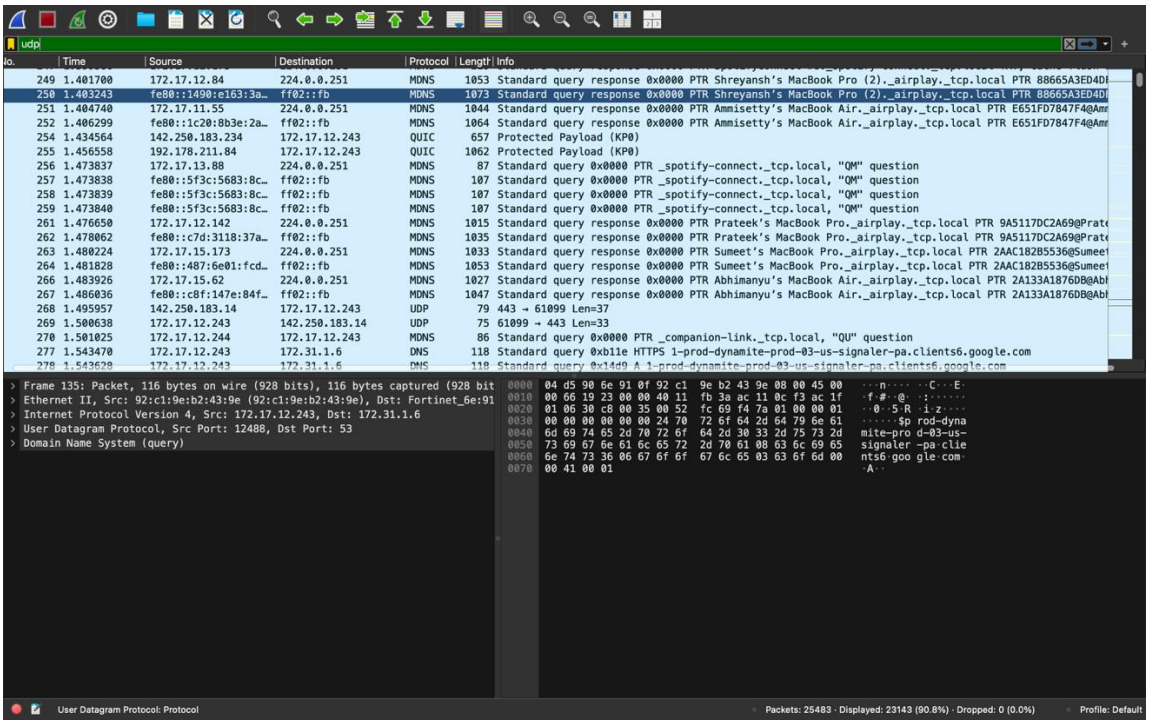
Internet Control Message Protocol: Protocol Packets: 10171 · Displayed: 106 (1.0%) Profile: Default

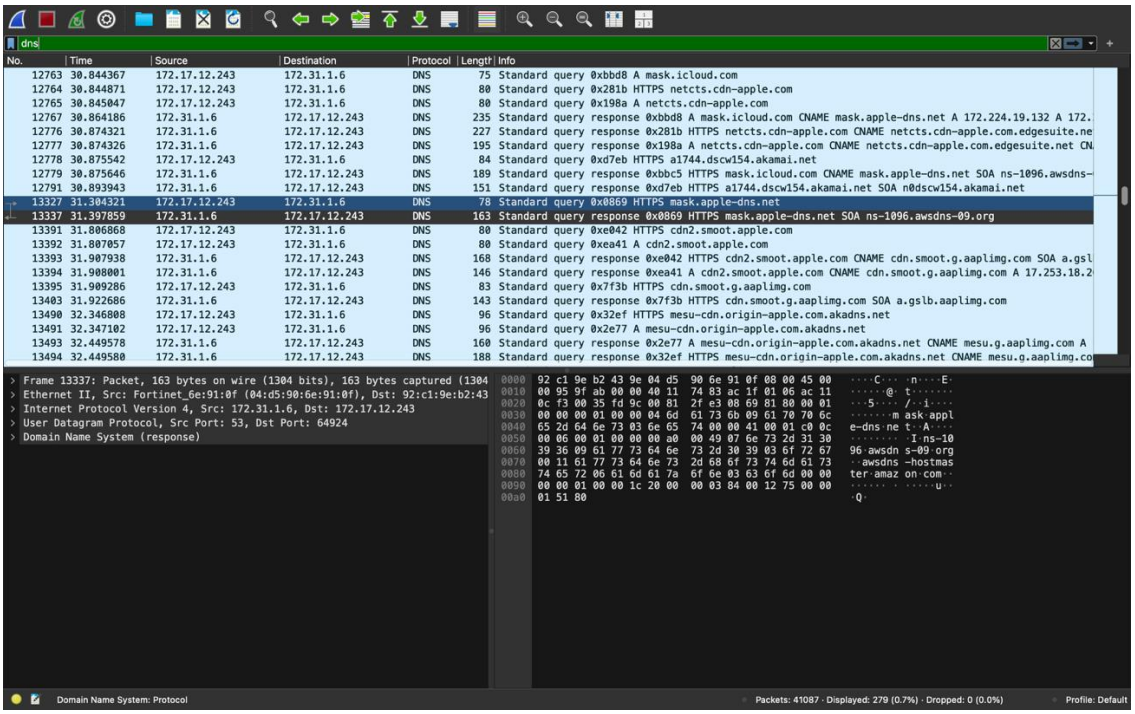
2. To analyze TCP packets and observe the TCP three-way handshak



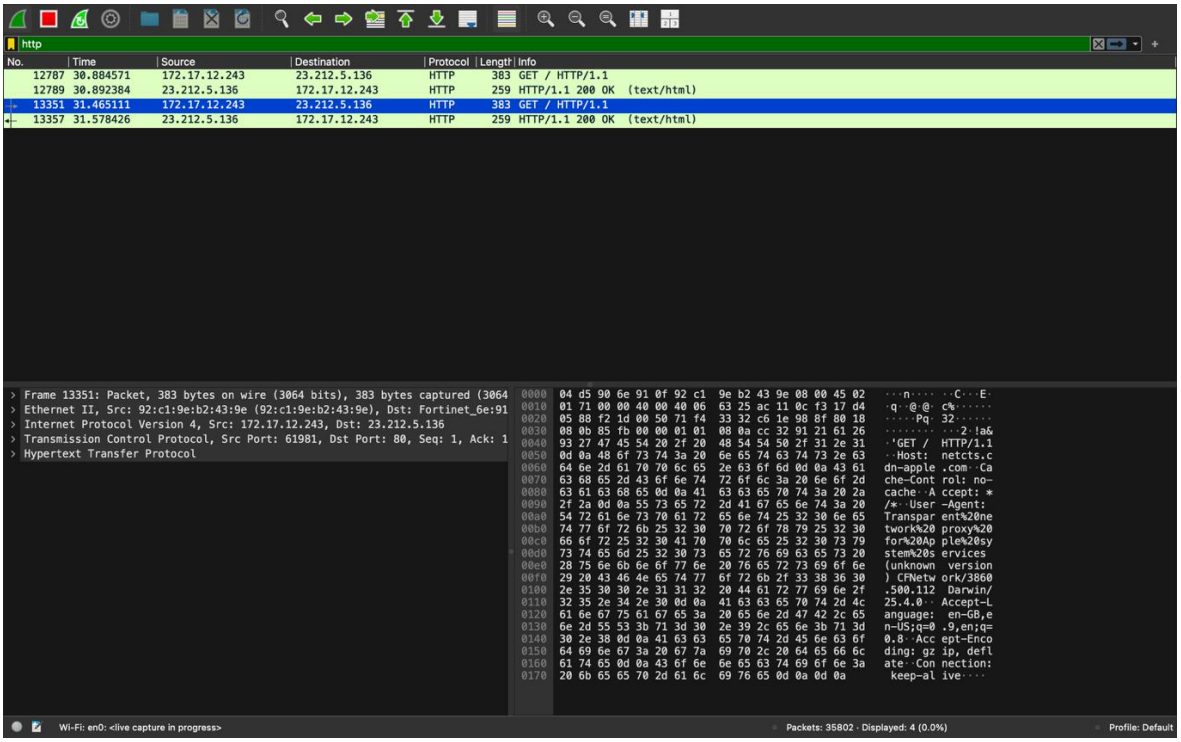
3. To capture and analyze UDP packets



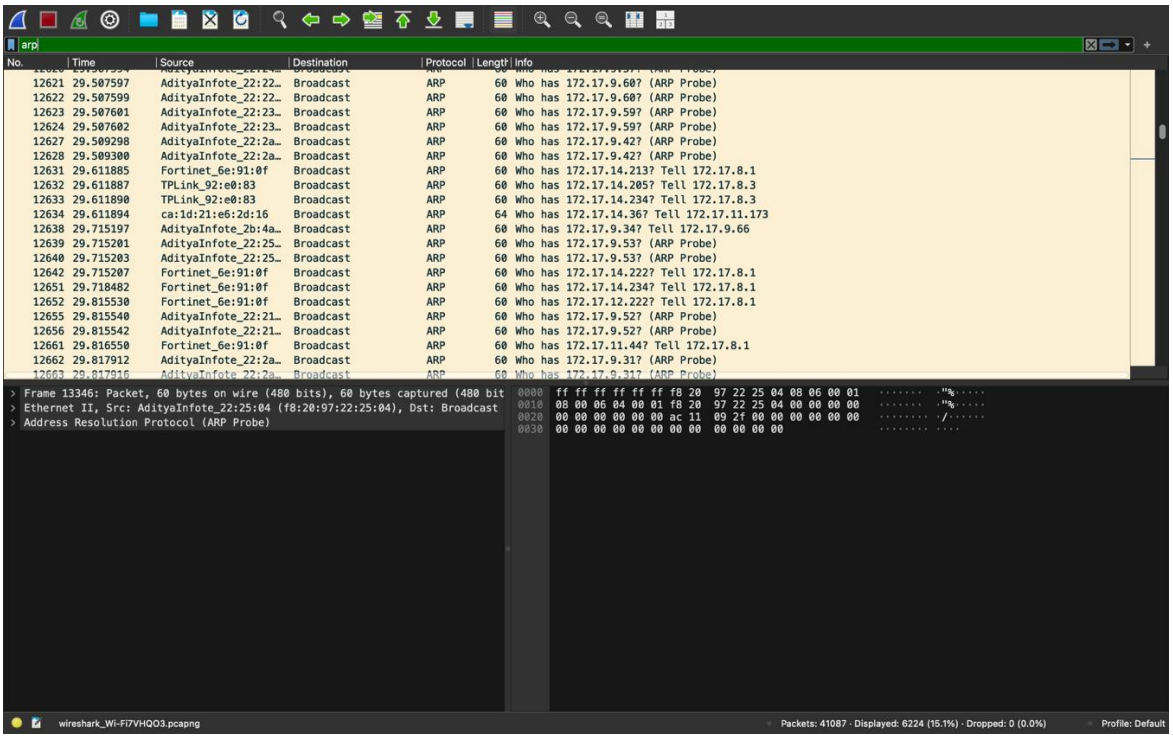
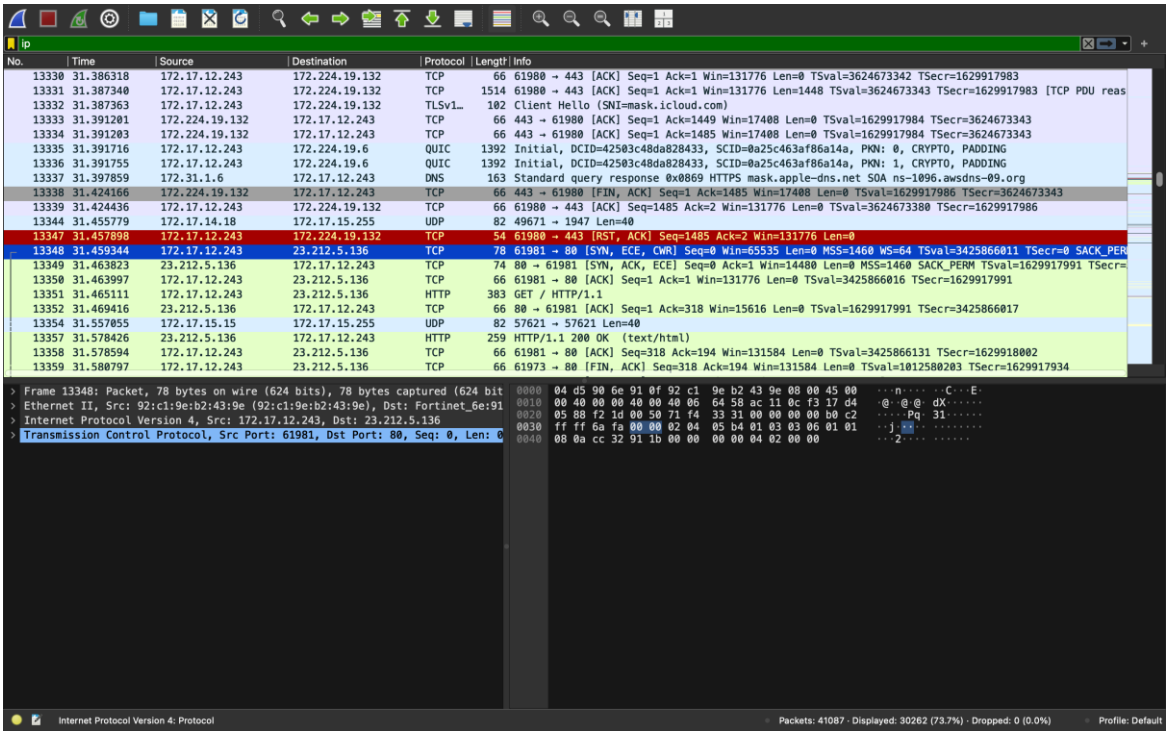
4. To capture and analyze DNS query and response packets



5. To capture and analyze HTTP packets



6. To analyze ARP request and reply packets



7. To identify source and destination MAC and IP addresses in captured packets